

# ISAIAH RAMIREZ

Computer Engineering Student | Hardware, Electronics & Robotics

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## TECHNICAL SKILLS

**Hardware & Electronics:** PCB design and layout, soldering, circuit assembly, schematic capture, analog/digital circuit analysis, hardware integration, troubleshooting

**Tools & Test Equipment:** KiCad, LTspice, Vivado, oscilloscope, function generator, multimeter, CAD modeling, 3D printing

**Digital & Embedded:** Verilog HDL, FPGA development, combinational/sequential logic, Arduino, microprocessors, assembly language

**Robotics & Programming:** Universal Robots, KUKA industrial robots, C/C++, Python, Java, Git, Linux

## RESEARCH EXPERIENCE

**UC Merced - Mi3 Lab | Undergraduate Research Assistant**

Summer 2026

*Autonomous Driving Research - Dr. Ross Greer*

- Prepared the RoboRacer autonomous vehicle for competition by supporting hardware integration, system readiness, and pre-deployment checks.
- Annotated natural-language driving instructions for the doPlan long-horizon autonomous-driving dataset, supporting vision-language planning research.
- Designed and built a custom power harness for an NVIDIA DRIVE AGX Thor developer kit to support future bench testing and hardware development.

**CSU Bakersfield | Undergraduate Research Participant**

Winter 2023

- Designed and built an Arduino-based automated plant-care prototype that measured humidity, displayed time, and activated watering at scheduled intervals.
- Modeled and fabricated custom components with CAD and 3D printing, integrating sensors, electronics, and mechanical parts into a functional system.

## ENGINEERING PROJECTS

**Stereo LM386 Audio Amplifier PCB**

LTspice | KiCad | Soldering

- Designed and simulated a stereo bridge-amplifier circuit in LTspice, then recreated the schematic and selected component footprints in KiCad.
- Completed PCB placement, trace routing, design-rule checking, and 3D verification; hand-soldered all through-hole components on the fabricated board.
- Validated operation with a function generator and laboratory equipment, producing variable-frequency audio output with adjustable volume.

**4-Bit FPGA Arithmetic Logic Unit**

Verilog | Vivado | Basys 3

- Designed a modular 4-bit ALU supporting AND, OR, addition, subtraction, and signed set-on-less-than operations with zero and overflow flags.
- Developed Verilog testbenches and verified waveforms before synthesis; mapped operands and controls to Basys 3 switches and results to board LEDs.

## EDUCATION & ROBOTICS TRAINING

**San Jose State University | B.S. Computer Engineering**

Expected 2027

**Relevant Coursework:** Electronics for Computing Systems, Digital Design I & II, Microprocessor Design I, Assembly Language, Algorithms & Data Structures, Software Engineering

**Bakersfield College | A.S. Computer Science; A.S. Physics**

2025

**Career Technical Education Center | Robotics Engineering Program**

2021

- Programmed and operated Universal Robots collaborative arms and gained hands-on experience with KUKA industrial robots, robotic workcells, and safety procedures.
- Earned Universal Robots and Robotics Engineering certificates.

## ADDITIONAL EMPLOYMENT

**Bolthouse Farms - Quality Assurance Technician**

Summer 2025

**Stria LLC / Bitwise Industries - Document Preparation Specialist**

2021-2023